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Background Information

Water policy significantly shapes both the management of San Antonio's water resources and the limitations of the data used in this study. Groundwater withdrawals from the Edwards Aquifer are regulated by the Edwards Aquifer Authority (EAA), but detailed pumping and recharge data especially from Aquifer Storage and Recovery (ASR) projects are not always publicly available or included in hydrologic datasets. This creates gaps in understanding how much observed aquifer fluctuation is due to climate versus human intervention.

Methodology of Data Analysis

We began by focusing on the climate parameters of average temperature and precipitation as explanatory variables. Water elevation in the J17 Index Well was chosen as the primary response variable due to the local importance of the aquifer as a regional water source. In addition to climate data, population data of the immediate region was added to assess well usage as a factor of aquifer recharge. We adopted the date range of 01-01-1991 to 12-31-2020 currently used by NOAA in climate normals for our analysis.

Data was obtained from open-source platforms (see References) and selected parameters were read into R. R-Studio was used to calculate and plot monthly and yearly moving averages and time series plots. Various packages

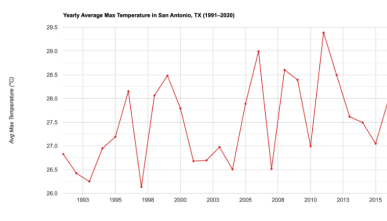
including `astsa`, `tseries`, `dynlm`, and `vars` were implemented in the analysis.

Exploratory Data Analysis

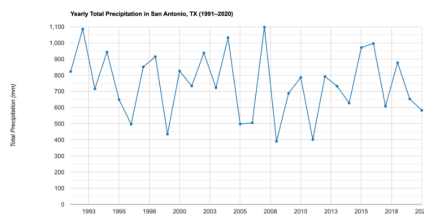
Daily temperature has clear cyclic behavior, with an apparent increase over time. Initial observations of yearly precipitation show a potential seasonal trend with possible increase in time between extreme measures. The water level in J17 appears to be spending more time at lower elevations in the latter portion of the time period studied. Population is steadily increasing over time, with a change in slope at or near 2000 (see Figure 1 for initial time series plots).

Figure 1. Initial time series plots

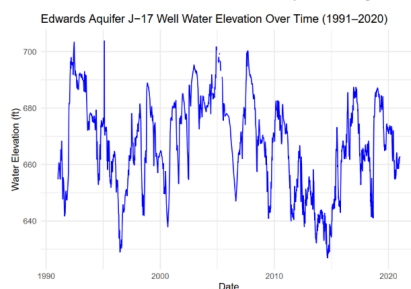
A. Temperature - monthly average



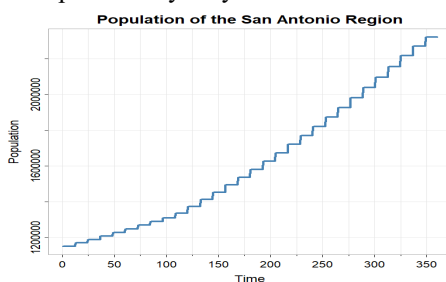
B Precipitation - monthly accumulation



C. Groundwater Elevation - monthly averages



D. Population - yearly census data



Handling Missing Data:

Upon further evaluation of the data, it became apparent that many of the datasets either contained missing data or were mismatched in their collection. For instance, the J17 well data is collected bi-monthly; the water elevation (the vertical measurement of the top of the aquifer from sea-level) is taken on the 15th and 30th of each month. Average temperature and precipitation data is taken daily. To adjust for this, it was decided to take a monthly average for temperature was calculated.. A simple mean was calculated for average temperature and J17 water elevation.

$$= \frac{1}{m} \sum_{t=1}^m x_t \quad m = \text{number of observations per month}$$

Given the nature of precipitation in San Antonio, a running monthly precipitation accumulation was taken instead. Using:

$$= \sum_{t=1}^m x_t \quad m = \text{number of observations per month}$$

In other cases, such as the population data mentioned above, the data was assumed to be constant for each year measured. This allowed for the use of a new data set concerning monthly data as opposed to daily data.

The J17 water elevation contained missing information at various times between July 2004 and August 2005. To account for this, a single value mean imputation was taken.

$$x_t = \frac{1}{n} (x_{t-2} + x_{t-1} + x_{t+1} + x_{t+2}) \quad n = \text{number of non - missing data}$$

The choice of using two values below and two above was made to account for consecutive, missing data.

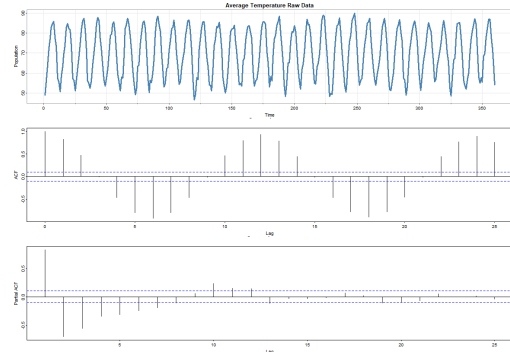
Results

Temperature Time Series Model

Assessments with ACF and Dickey-Fuller tests revealed that the temperature data was already in a stationary form. When taking a slight positive trend of the data into account, Dickey-Fuller p-values are below 0.01 at all lags (confirming stationarity).

Preliminary analysis of the ACF and PACF (Figure 2) indicate a potential Seasonal AR(1), MA(1), and non-seasonal AR(1).

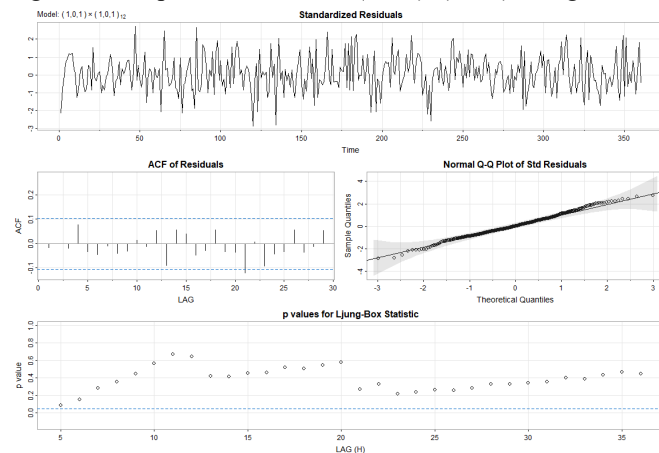
Figure 2. Temperature time series plot, ACF, and PACF



The final model decided upon was SARIMA(1,0,1) $\times$ (1,0,1)₁₂ with the following coefficients and standard errors:

$$x_t = 70.037_{(4.370)} + 0.601_{(0.119)} x_{t-1} + 1.000_{(0.0001)} x_{t-12} - 0.340_{(0.141)} w_{t-1} - 0.975_{(0.041)} w_{t-12} + w_t$$

While several models with and without seasonality were tested, this was determined to be the strongest model based on investigation of the residuals and AIC/BIC calculations for the given parameters. Our chosen model had the 2nd lowest AIC/BIC, but Q-statistics indicated that this model produced the most white noise residuals of any model, as shown below (Figure 3).

Figure 3. Temperature SARIMA(1,0,1) $\times$ (1,0,1)₁₂ diagnostics

Temperature forecasting with this model produced plausible values for the following two years, as seen in Figure 4, below. Additionally, this model was confirmed by plotting the actual monthly average temperatures for 2021-2022 alongside values forecasted by the chosen SARIMA. Actual temperatures were within confidence intervals, and reinforced the choice of SARIMA(1,0,1) $\times$ (1,0,1)₁₂ model (Figures 4 and 5).

After confirming the appropriate model for temperature, it was important to check for nonconstant variance to see if an

ARCH/GARCH model needed to be applied to the series. Nonconstant variance was not suspected, however, due to the highly cyclic nature of this series. Examining the P/ACF of the residuals (Figure 6) and squared residuals (Figure 7) indicate no significant lags in either case. No ARCH behavior is present in the temperature series.

Figure 4. Temperature forecasting for 2021-2022.

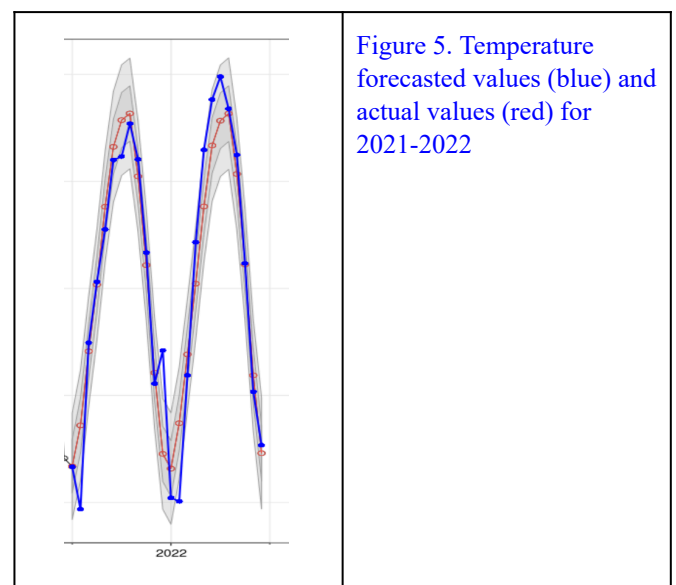
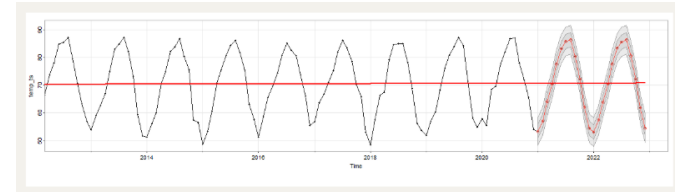


Figure 5. Temperature forecasted values (blue) and actual values (red) for 2021-2022

Figure 6. Residual P/ACF for temperature data

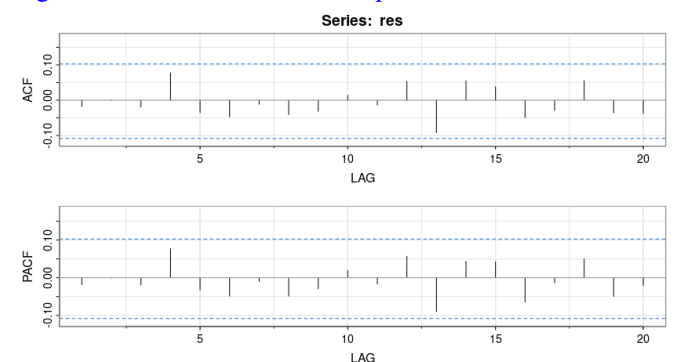
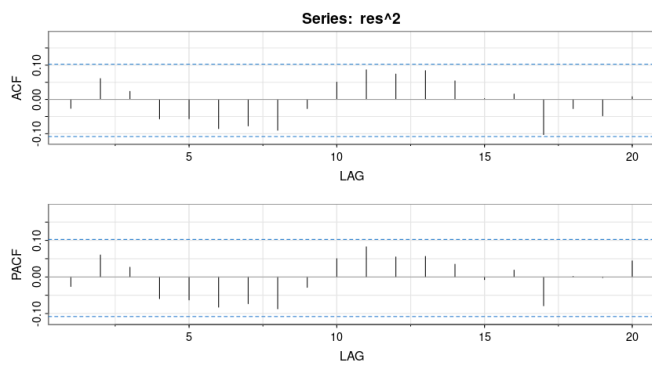


Figure 7. Squared Residual P/ACF for temperature data

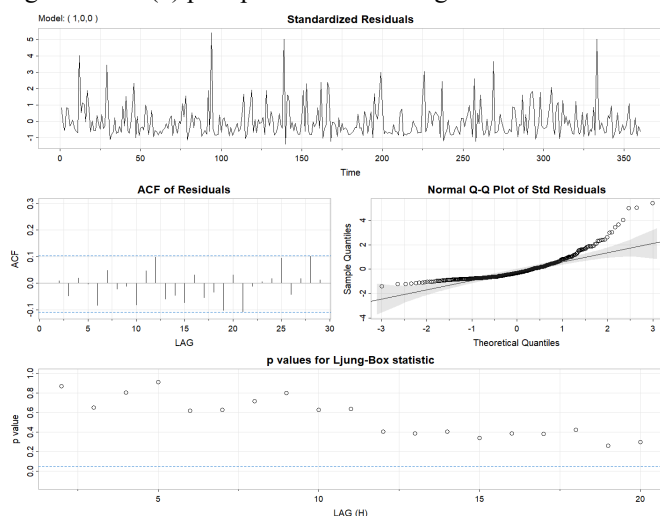


Precipitation Time Series Model

Early assessments using Autocorrelation Function (ACF) plots and the Augmented Dickey-Fuller (ADF) test revealed that the precipitation data was already stationary, requiring no further transformation for time series analysis. A log transformation was attempted after using the Box-Cox function in R, but residuals were not noticeably improved.

Analysis of the precipitation data's ACF and PACF graphs suggested three possible models: AR(1), ARIMA(1,1), and a seasonal AR(1). Diagnostic residual plots suggest that AR(1) is the strongest model for precipitation (Figure 8); while the Q-Q plot shows a significant amount of deviation from a normal distribution, it is still an improved model when compared to the other potential models. We suspect these deviations may be due to potential dependence upon another variable, and will investigate this relationship during the next phase of our project.

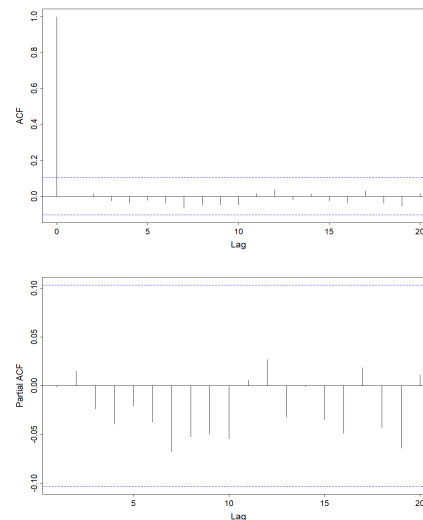
Figure 8. AR(1) precipitation model diagnostics



As both the original data (Figure 1B) and AR(1) residuals (Figure 8) indicate slight possible heteroscedasticity, the necessity of including a GARCH model was investigated.

However, after the ACF and PACF (Figure 9) of the squared residuals from the AR(1) model were further investigated, no peaks were observed in either ACF nor PACF, suggesting that variances are not changing significantly over time, and a GARCH model was not deemed necessary.

Figure 9. ACF and PACF of Precipitation AR(1) squared residuals



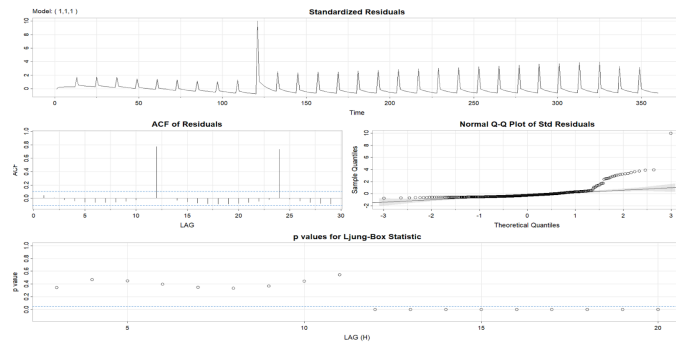
For completeness, a GARCH(1,1) model was overfit to the model to confirm that these parameters were unnecessary.

Population Data Model

The raw population data did not appear to exhibit stationarity due to a mean value clearly dependent on time. The data was log transformed to stabilize variance. Stationarity was achieved first through differencing logged data. A second attempt was made by detrending the logged data using a linear regression trend model. Although a log transformation was applied to stabilize variance, and first-order differencing and linear detrending were used in an attempt to remove the trend and achieve stationarity, these transformations were insufficient.

While the ACF and PACF plots suggested the potential for AR(1), MA(1), or ARMA(1,1) models, the residuals from the ARIMA models did not meet the key assumptions of white noise. Specifically, the residuals were not normally distributed and exhibited significant autocorrelation, as evidenced by Q-statistics with p-values less than 0.05. This indicates that the model failed to capture key patterns in the data, leaving structure in the residuals and rendering the model statistically invalid. A representative diagnostic of one of these attempted models (ARIMA(1,1,1)) may be seen in Figure 10 on the following page. Of note are the clearly non-normally distributed residuals and Q-statistics not greater than the conventional $p=0.05$, indicating a non-white noise residuals.

Figure 10. Diagnostics plot of representative ARMA models for population data

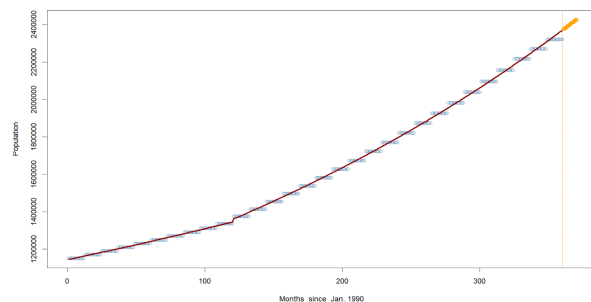


Subsequently, a simple linear regression model was then fit to the logged population data. This resulted in the model:

$$\log(\text{Population}) = 13.95_{(0.0014)} - 0.1062983_{(0.0022)}D + 0.0013479_{(0.00002)}t + 0.0009709_{(0.00002)}D*t,$$

where D is a dummy variable (0 before Jan. 2000 and 1 from Jan. 2000 onwards). All coefficients were significant ($p < 0.001$), and $R^2 = 0.9989$. This model was aligned with the observed data from Jan. 1990 to Dec. 2020 as well as used to forecast ten years of population data (see Figure 11, below).

Figure 11. Population data, modeled values, and predicted values vs. months since Jan. 1990



Furthermore, ARIMA models are univariate and do not account for external influences such as zoning policies, economic development, infrastructure projects, or social changes, all of which play significant roles in shaping population trends. Because of these limitations, ARIMA is not well-suited for modeling population time series data on its own. A more appropriate approach was to treat population as a deterministic variable modeled using linear growth function

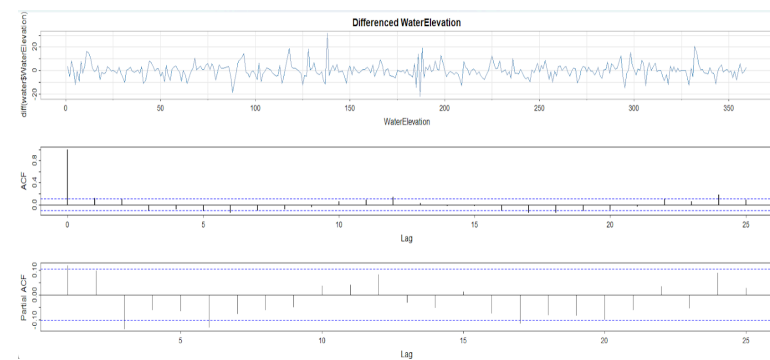
As population was assumed to be constant through each year and ultimately fit to a simple linear regression model

based on time, incorporating a variance model such as ARCH or GARCH is not appropriate. Future studies that are able to incorporate more exact population numbers for the area would potentially add an additional level of accuracy to this model.

Well Water Elevation Time Series Model

The J17 well water elevation time series was coerced to stationary by performing a simple first order differencing of the data. The transformed data can be seen below (Figure 12). It is obvious from the differenced data that the ACF converges to zero quickly, has a constant mean, and has a relatively uniform variance, indicating stationarity of the transformed data. The PACF plot has few spikes after a lag of 3, suggesting a potential ARIMA(3,1,0) model.

Fig. 12 Differenced well water elevation time series, ACF, and PACF plots



While the ARIMA(3,1,0) model had white noise, normally distributed residuals, several other models were also analyzed to ensure selection of the strongest model. Other ARIMA orders tested were (2,1,0), (1,1,0), and (3,0,0). Interestingly, the ARIMA(3,0,0) not only resulted in appropriate residuals, it also had reduced AIC and BIC measures. The ARIMA(3,0,0) model was selected for the final model due to the strong fit and parsimonious nature of the model:

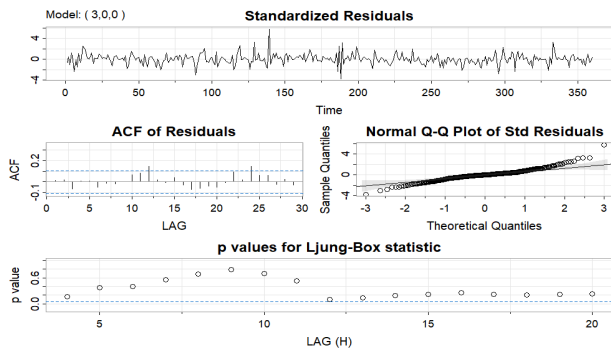
$$(x_t - 667.2881) = 1.051(x_{t-1} - 667.2881)_{(0.0520)} - .1495(x_{t-3} - 667.2881)_{(0.0520)}$$

The model could be further reduced by removing the statistically insignificant lag(2) term. With a coefficient of .0036 and a p-value of .9628, the term was deemed no different from zero and thereby unnecessary for the model. The non-differenced model is not centred around zero, so a mean correction was also introduced into the model.

Further diagnostics for this model are presented in Figure 13. Here, it is seen that standardized residuals appear to be stationary and relatively normally distributed. Ljung-Box Q

statistics do not provide sufficient evidence to reject the null hypothesis of white noise residuals.

Figure 13. Diagnostics of Groundwater Elevation ARIMA(3,0,0)



The model was fit against the previously observed data and exhibited a very strong fit to the original data, as seen in Figure 14, below. Additionally, forecasted values produced were compared to the actual 2021 water elevation data (Figure 15). The forecasted data fit well with the true water elevation. While forecasted values are conservative, the model reflected the general change in water elevations well.

Figure 14. Groundwater elevation model vs observed values

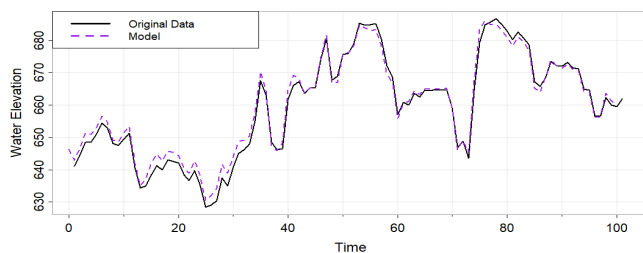
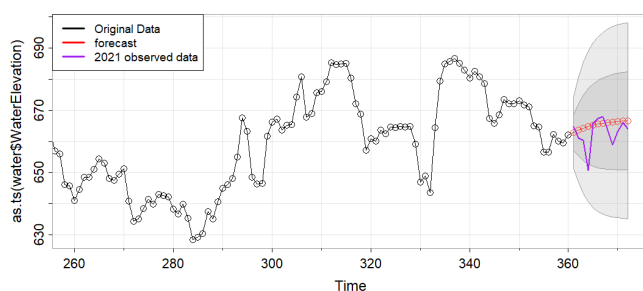


Figure 15. Groundwater 2021 forecasted values

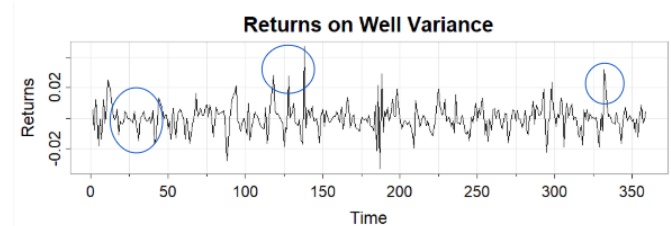


Volatility in the Water Elevation with ARCH:

Upon further evaluation of the water elevation data, it was deemed that the volatility may also change over time. The standardized residuals exhibit some leftover correlation, albeit small (Figure 16). We can also see some inconsistent

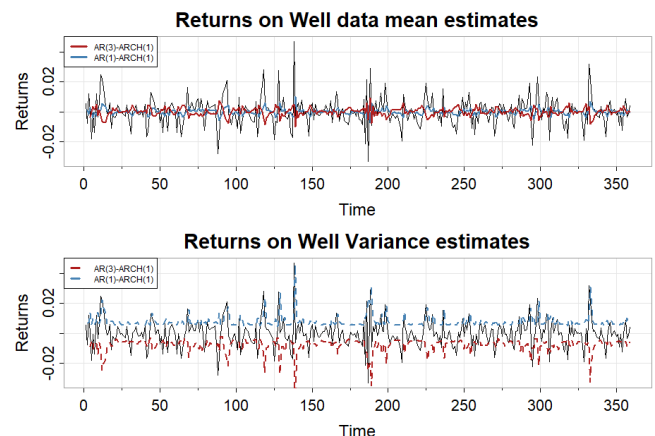
variation within the original time series plot (Figure 1 C), and the returns of the stationary (differenced and log transformed) data reveal small changes in variation across time (Figure 16). This indicates that the data may exhibit heteroscedasticity.

Figure 16. Log Returned Well Data



To account for this possible heteroscedasticity, two ARCH models were built to estimate these changes in volatility. The ACF and PACF plot of the returned residuals on (ARIMA(3,0,0)) cut off after lag(1) for both the ACF and PACF. However, when working with the GARCH(1,1) model, in no instances where the omega or beta terms found to be significant. Therefore the models AR(3)-ARCH(1) and AR(1)-ARCH(1) were analyzed. In (Figure 17) we can see a comparison between the two options. (Note: the Variance of AR(3)-ARCH(1) has been inverted to allow for easier comparison).

Figure 17. ARCH(1) fits for Well data on means and variances



In both cases the variation fits the observed data quite well, and both exhibit lower AICs than the original model being -6.69 and -6.71 (AR(1) and AR(3), respectively). However, it appears that the AR(3) model seems to be inverted from the original data in the returned mean estimates, and exhibits non-significant AR(2) and AR(3) terms. Therefore, the chosen model was an AR(1)-ARCH(1) model as seen below.

$$r_t = .2118_{(.06014)}x_{(t-1)} + \sigma_t \varepsilon_t$$

$$\sigma_t^2 = .00004650_{(.000005297)} + .5198_{(.1281)} r^2_{(t-1)}$$

The AR(1)-ARCH(1) model was then re-fit against the observed data. This model had a similar fit for the original data as did the original chosen model. As the model was constructed under the log returned data, it was necessary to back transform the mean estimate for forecasting using:

$$r_t = \log(x_t) - \log(x_{t-1})$$

$$e^{(r_t + \log(x_{t-1}))} = x_t$$

No centering adjustment was needed as the mean of the returned data was $.00002258 \approx 0$ and exhibited a t-statistic of $-.048$ with a p-value of $.962$. The forecast was then compared to the actual 2021 water elevation data in both the mean forecast and the volatility (Figures 18 and 19).

Figure 18. Forecasting means

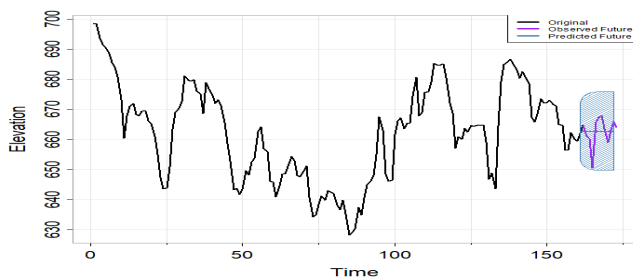
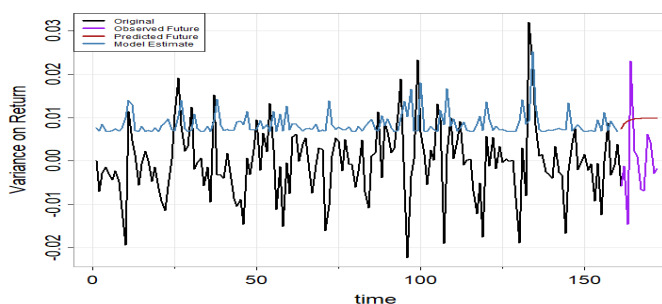


Figure 19. Forecasting volatility



The new model's forecast is less optimistic than the original model's forecast; however, it seems to fit the mean of the future data better. The volatility also seems to match the future data well, with an expected standard deviation in log returns of 1.5% before increasing to 2.0% in the year of 2021.

Multivariate Model Analysis

Several simple linear regression models were analyzed for the well water elevation response variable. All models

included average monthly temperature and monthly accumulated precipitation as predictor variables. These two predictor variables produced statistically significant coefficients in every tested model. Population and various interaction terms were also tested for significance, but were not found to be significant in any simple linear regression model, and were thus excluded from future multivariate analyses. The cross correlation function of temperature and precipitation related to well water elevation was computed and a lag in water elevation of 1 month was determined to have the highest correlation to both predictor variables. Thus, a model including lagging well water elevation on predictor variables was also analyzed. Model strength was determined by analyzing the coefficient of determination, AIC, and BIC of each respective model. Results are presented in Table 1.

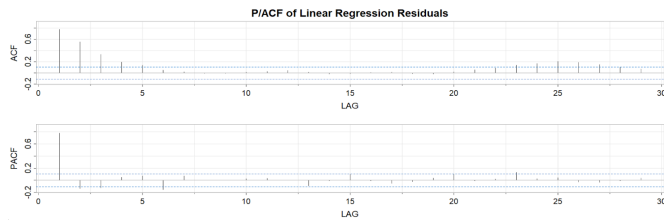
Table 1. Simple multiple linear regression model results

MODEL	Adjusted R ²	AIC	BIC
Well = Precipitation + Temperature + Population	0.2216	2111.901	2131.119
Well = Precipitation + Temperature	0.2235	2110.091	2125.465
Well = Precipitation + Temperature + Precipitation*Temperature	0.2229	2111.004	2130.222
Well = Leading Temperature + Leading Precipitation	0.2235	2226.716	2242.25

The strongest analyzed simple linear regression model was determined to also be the most parsimonious model, with well water elevation predicted by precipitation and temperature alone.

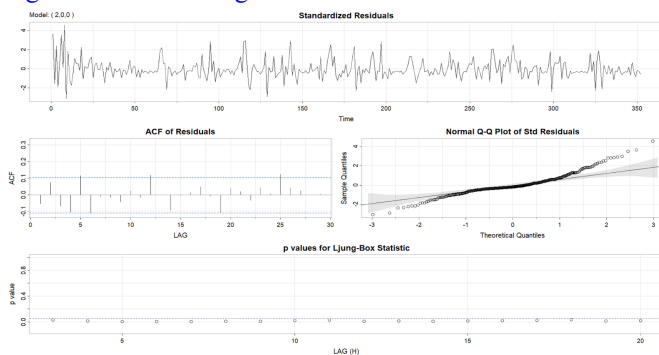
It must be noted that this model only captured ~22.4% of the variation in well water elevation by the predictor variables. This model indicates a moderate positive linear correlation between well water elevation and precipitation and a moderate negative linear correlation between well water elevation and temperature, both of which are logical results. The residuals of the model were then investigated for potential autocorrelation (Figure 20). The ACF of the residuals from this model decays to near-zero, while the PACF has strong peaks at 1-2 lags.

Figure 20. P/ACF of Linear Regression Residuals



Residuals were fit to both an AR(1) and AR(2) to test for autocorrelation of errors. Of these two, AR(2) was a marginally better fit, though neither model produced white noise, normally distributed residuals (Figure 21). The residuals of the autocorrelated model still exhibited potential changes in the variation, though no trend was observed. Residuals had heavier than normal tails, and Q statistics indicated substantial evidence against white noise residuals.

Figure 21. Model Diagnostics for Autocorrelated Errors



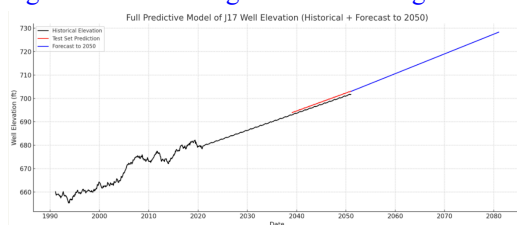
Including modeled autocorrelated errors in the linear regression model was not determined to benefit the fit or strength of the model.

While the simple linear regression of

$$\text{WellWaterElevation} = 0.33816_{(0.28188)} \text{Precipitation} - 0.37378_{(0.04093)} \text{Temperature}$$

was found to be the best possible linear regression model, this model does not appear to be a particularly strong fit to the data. Moreover, the assumption of independent and identically distributed errors is frequently violated in groundwater datasets, where residuals often exhibit volatility clustering and lags patterns better suited for time series models like ARIMA or volatility models like GARCH. Linear regression also fails to incorporate feedback loops or delayed effects, such as the lag between recharge events and actual changes in aquifer levels.

Figure 22. Linear Regression Modeling Forecast

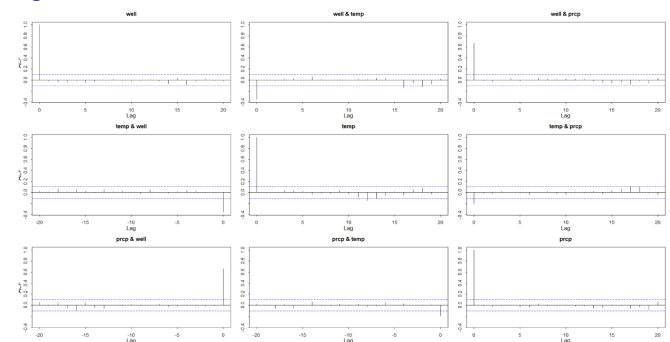


Such as this example of the well elevation, forecasting is not as linear as it should be. As a result, while linear regression can suggest general trends, it oversimplifies the system and may lead to misleading conclusions or underestimation of risk in water resource planning. More advanced models that account for temporal dependence, nonlinearity, and external shocks are typically more appropriate for reliable forecasting in such complex environmental systems.

To improve the fit of a regression model, a multivariate time series regression model was analyzed. This is appropriate after determining that the response variable (well water elevation) and both significant predictor variables (temperature and precipitation) are all separate time series.

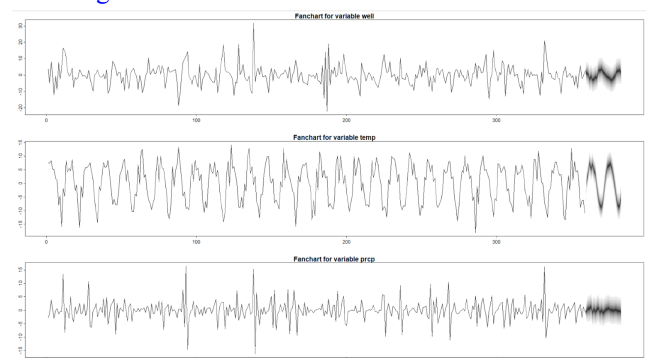
The vars package in R was implemented to assist with the complicated nature of this model. This algorithm calculated a selected lag of 12 months. With the inclusion of this lag as well as all three time series analyzed simultaneously, the coefficient of determination was increased to 0.3503, a 53.4% increase in the proportion of variability in well water elevation accounted for by the predictors. Analysis of the residuals of this multivariate time series regression model indicate no significant autocorrelation of errors (Figure 23).

Figure 23. ACF of multivariate time series model residuals



Forecasts using the multivariate time series regression model provide estimates in agreement with observed values (Figure 24).

Figure 24. Predictions for 2021-2022 from multivariate time series regression model.



Conclusion and Future Analysis

This investigation explored the potential impact of climate change on municipal water sources in San Antonio, Texas, over the period from 1991 to 2020. By analyzing time series data on climate variables, groundwater, and population growth, we identified observed and predictive models for each variable.

This study provides an in-depth examination of the relationship between climate variability, population growth, and groundwater availability in San Antonio, Texas, through the lens of advanced time series modeling. By applying both univariate and multivariate statistical approaches, including ARIMA, SARIMA, ARCH, and GARCH models, we gained valuable insights into the behavior of temperature, precipitation, population growth, and aquifer response over the period from 1991 to 2020. Our findings suggest that while temperature and precipitation each exhibit distinct trends and seasonal patterns, their combined influence on groundwater elevation is moderate at best in simple regression models but significantly enhanced when incorporated into multivariate time series frameworks. The multivariate time series model, which captured nearly 36% of the variance in aquifer elevation, outperformed linear regression and demonstrated the importance of considering lagged, temporally structured interactions among variables.

Temperature was shown to have a final SARIMA $(1,0,1) \times (1,0,1)_{12}$ model, while precipitation and well water elevation appear to follow autoregressive models of order 1 and 3, respectively. Population growth followed a simple linear regression based on time. Diagnostics of each model confirmed the validity of each model by producing normally distributed white noise residuals.

Ongoing analysis with more advanced modeling techniques may offer deeper insights and inform mitigation strategies to ensure long-term water sustainability in the face of climate change.

Additional Future Analysis

The final simple linear regression model was analyzed for autocorrelated errors, but it was not able to additionally include the GARCH model fit to the well water elevation time series. In the future, including both the mean and variance models for the response variable would likely improve the fit and predictive power of this model, though it likely will still not be as precise and accurate as the multivariate time series model.

Interestingly, the automatic lag computation for the multivariate time series analysis suggested a lag of 12 months rather than the one month seen with the highest peak

in the cross correlation function plots between separate time series. This is likely due to the algorithm used with multiple time series.

Not surprisingly, the coefficient of determination of the multivariate time series model was much higher than that of the simple linear regression between well water elevation predicted by temperature and precipitation. Again, the multivariate time series model utilizes a more complete picture of all information in its ability to incorporate the time series components of all three series together. While the R^2 was still not considered “large” at 0.3503, it is double that of the simple linear regression R^2 of 0.241. This highlights the need for not disregarding the time series component of any variable when it is available.

Modern Technological Complications In Water Resource Analysis

One of the most complex challenges in evaluating the impact of climate change on municipal water supplies in San Antonio stems from modern water management practices and infrastructures that are not directly captured in standard environmental datasets. While precipitation, temperature, and aquifer elevation can reflect some natural responses to climate variability, they often fail to isolate or document human-induced changes like artificial recharge, groundwater pumping, and inter-basin transfers.

A prime example is Aquifer Storage and Recovery (ASR), a process whereby excess surface water often drawn from the Edwards Aquifer during wet periods is injected into confined zones of the aquifer for later use. These injection volumes can raise groundwater levels even during drought years, giving a false impression of natural recharge. Unfortunately, datasets from USGS and TWDB typically report groundwater elevation without distinguishing between natural recharge (e.g., infiltration from rainfall) and managed recharge via ASR. As a result, interpretations of aquifer response to climate variability can be confounded by unknown levels of engineered intervention.

Similarly, municipal and agricultural groundwater pumping which varies seasonally and with population growth, irrigation demand, and industrial activity is not recorded in the daily or monthly groundwater level datasets used in this study. Large-scale pumping during dry summers or drought emergencies can lower aquifer levels even if precipitation remains steady. Without concurrent pumping withdrawal data, declines in aquifer elevation may be wrongly attributed solely to climate factors.

Another major limitation involves evapotranspiration (ET). As temperatures rise, ET increases, drawing more water from both soil and surface reservoirs. However, accurate ET estimates are often derived from remote sensing tools (e.g., MODIS, SEBAL, METRIC) or weather

station-based models (e.g., Penman-Monteith), and such estimates were not incorporated into this study. Without ET data, it is difficult to fully assess how much of San Antonio's rainfall is actually contributing to aquifer recharge versus being lost to atmospheric return. In a semi-arid climate like Central Texas, where small shifts in ET can tip the balance between recharge and deficit, ignoring ET underestimates water loss mechanisms critical to sustainability planning.

Compounding these issues are engineered modifications to surface water systems, such as dam releases, water transfers between regions (e.g., Guadalupe-Blanco River Authority actions), and changes in urban drainage systems. These human-controlled flows are not captured in raw "percent full" readings of Canyon Lake, leading to misinterpretations about whether reservoir declines are due to decreased inflow (from rainfall) or increased withdrawals and releases.

Lastly, data reporting and agency silos make integration difficult. Climate data may come from NOAA, water level data from the Edwards Aquifer Authority, and population data from the U.S. Census all with different resolutions, update frequencies, and metadata standards. Lack of centralized, harmonized hydrologic databases creates analytical blind spots in system-level understanding.

To address these limitations, future studies should incorporate a more holistic water balance approach, including estimates of ASR volumes, pumping rates, and evapotranspiration. Additionally, advanced hydrologic models (e.g., MODFLOW coupled with water-use inputs or remote-sensing-based ET) should be employed to distinguish natural from anthropogenic signals in aquifer and reservoir trends. Only by accounting for these complexities can we fully understand the water resource vulnerabilities of San Antonio in an era of rapid climate change and urban growth.

Limitations of Data

Despite the insights gained from this study, several critical limitations in the data collection and processing must be acknowledged, as they constrain the interpretation and generalizability of the findings.

First, the inconsistency in temporal resolution between datasets introduces substantial challenges. Daily precipitation and temperature data from NOAA are highly granular, whereas the J17 Index Well is measured only twice per month (on the 15th and 30th). To harmonize these data for comparative analysis, all time series (save precipitation, which was included as monthly accumulation) were aggregated into monthly averages. However, this temporal smoothing can obscure high-impact hydrological events such as flash floods, rapid recharge pulses, or short-term pumping surges events that could significantly influence aquifer response but are diluted in monthly means.

Second, missing or incomplete data reduced the continuity and robustness of the analysis. For example, the J17 well data includes gaps between July 2004 and August 2005, a period that required imputation using a multi-point neighborhood average (i.e., averaging values two months before and after). While this method is a reasonable statistical estimate for limited gaps, it assumes that the aquifer follows predictable trends, which may not be valid during periods of drought, flooding, or heavy pumping. Likewise, historic population data was not available at monthly resolution, so a steady yearly population was assumed.

Third, the spatial scope of the analysis was limited by the availability of point-specific data. The J17 Index Well, located near Fort Sam Houston, is used as a regulatory and monitoring reference point for the San Antonio segment of the Edwards Aquifer. While valuable, it does not represent the entire aquifer system, which spans multiple recharge zones, spring complexes, and confining units extending through Comal, Bexar, Hays, and Medina counties. The localized nature of these datasets introduces spatial bias and limits our ability to capture region-wide aquifer dynamics, lateral recharge variability, or inter-aquifer transfers.

Finally, while the study's 30-year duration (1991–2020) provides a long enough record to detect interannual trends and decadal oscillations, long-term climate-hydrology interactions such as those tied to ENSO (El Niño–Southern Oscillation), Pacific Decadal Oscillation, or anthropogenic climate trends may operate over longer scales. The limited number of high-intensity drought years (e.g., 2011, 2008) within this period also limits statistical robustness when evaluating extreme events. Additionally, changes in land cover, impervious surface expansion, and shifts in vegetation across San Antonio's watershed were not accounted for, though they can substantially influence infiltration, runoff, and evapotranspiration rates.

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